

1(a)

The variable X has the distribution $B(50, \frac{1}{6})$. The probabilities $P(X = r)$ for $r = 0$ to 50 are given by the terms of the expansion of $(a+b)^n$ for specific values of a , b and n .

State the values of a , b and n .

[1]

- (b) A student has an ordinary 6-sided dice. They suspect that it is biased so that it shows a 2 on fewer throws than it would if it were fair. In order to test the suspicion the dice is thrown 50 times and the number of 2s is noted. The student then carries out a hypothesis test at the 5% significance level.

Write down suitable hypotheses for the test.

[2]

- (c) Determine the rejection region for the test, showing the values of any relevant probabilities.

[4]

- 2 In the past, the time for Jeff's journey to work had mean 45.7 minutes and standard deviation 5.6 minutes. This year he is trying a new route. In order to test whether the new route has reduced his journey time, Jeff finds the mean time for a random sample of 30 journeys using the new route. He carries out a hypothesis test at the 2.5% significance level.

Jeff assumes that, for the new route, the journey time has a normal distribution with standard deviation 5.6 minutes.

State appropriate null and alternative hypotheses for the test.

[2]

- 3 Pierre is a chef. He claims that 90% of his customers are satisfied with his cooking. Yvette suspects that Pierre is over-confident about the level of satisfaction amongst his customers. She talks to a random sample of 15 of Pierre's customers, and finds that 11 customers say that they are satisfied. She then performs a hypothesis test.

Carry out the test at the 5% significance level.

[7]

- 4(a) Some packets of a certain kind of biscuit contain a free gift. The manufacturer claims that the proportion of packets containing a free gift is 1 in 4. Marisa suspects that this claim is not true, and that the true proportion is less than 1 in 4. She chooses 20 packets at random and finds that exactly 1 contains the free gift.

Use a binomial model to test the manufacturer's claim, at the 2.5% significance level.

[7]

- (b) The packets are packed in boxes, with each box containing 40 packets. Marisa chooses three boxes at random and finds that one box contains 19 packets with the free gift and the other two boxes contain no packets with the free gift.

Give a reason why this suggests that the binomial model used in part (a) may not be appropriate.

[1]

- 5 It is known that, under the standard treatment for a certain disease, 9.7% of patients with the disease experience side effects within one year.

In a trial of a new treatment, a random sample of 450 patients with this disease was selected and the number X who experienced side effects within one year was noted.

It was found that 51 of the 450 patients experienced side effects within one year.

Test, at the 10% significance level, whether the proportion of patients experiencing side effects within one year is greater under the new treatment than under the standard treatment.

[7]

6 Research has shown that drug A is effective in 32% of patients with a certain disease.

In a trial, drug B is given to a random sample of 1000 patients with the disease, and it is found that the drug is effective in 290 of these patients.

Test at the 2.5% significance level whether there is evidence that drug B is effective in a lower proportion of patients than drug A.

[7]

- 7 (a) André throws a fair six-sided dice 30 times. The number of throws on which the score is six is denoted by X .
- (i) State a suitable model for X , including the values of any parameters. [1]
- (ii) Find $P(X = 9)$. [1]
- (iii) Find $P(X \geq 9)$. [2]
- (b) André has another six-sided dice. He suspects that this dice is biased so that it is more likely to show a six than if it were fair. He throws the dice 30 times and it shows a six on 9 throws. Test at the 5% significance level whether André's suspicion is justified. [6]

- 8 It is known that under the standard treatment for a certain disease, 9.7% of patients with the disease experience side effects within one year. In a trial of a new treatment, 450 patients with this disease were selected and the number, X , that experienced side effects within one year was noted.

It was found that 51 of the 450 patients experienced side effects within one year.

- (a) Test, at the 10% significance level, whether the proportion of patients experiencing side effects within one year is greater under the new treatment than under the standard treatment.

[7]

- (b) It was later discovered that all 450 patients selected for the trial were treated in the same hospital. Comment on the validity of the model used in part (a).

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		$n = 50; a = \frac{5}{6}, b = \frac{1}{6}$ or $a = \frac{1}{6}, b = \frac{5}{6}$,	B1 [1]	or $\left(\frac{5}{6} + \frac{1}{6}\right)^{50}$ or $\left(\frac{1}{6} + \frac{5}{6}\right)^{50}$
	b		$H_0: p = \frac{1}{6}$ allow 0.167 or 0.17 $H_1: p < \frac{1}{6}$ $p = P(\text{dice shows a 2 (on any throw)})$ or $p = \text{proportion of 2s thrown}$	B1 B1 [2]	One error, eg undefined p or 2-tail or H_1 : $p > \frac{1}{6}$: B1B0 Two errors, eg $H_1: p > \frac{1}{6}$ and undefined p : B0B0 Cannot be scored in part (c)
	c		$B(50, \frac{1}{6})$ Attempt $P(X \leq a)$ for $1 \leq a \leq 20$ $P(X \leq 4) = 0.0643$ $P(X \leq 3) = 0.0238$ Rejection region is ≤ 3 twos or < 4 twos Alternative method ft if 12(b) $H_1: p > \frac{1}{6}$ $B(50, \frac{1}{6})$ Attempt $P(X \geq a)$ for $1 \leq a \leq 20$ $P(X \geq 14) = 0.0307$ $P(X \geq 13) = 0.0627$ Rejection region is ≥ 14 twos or > 13 twos	B1 M1 A1 A1 Alternative method B1 M1 A1 A1 [4]	stated or implied or $P(X < 5) = 0.0643$ or $P(X < 4) = 0.0238$ Must see both probabilities correct oe. Allow $X \leq 3$ or $X < 4$ Alternative method ft if 12(b) $H_1: p > \frac{1}{6}$ stated or implied Must see both probabilities correct oe. Allow $X \geq 14$ or $X > 13$
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2			$H_0: \mu = 45.7,$ where μ = mean of all new journey times $H_1: x < 45.7$	B1 (AO1.1) B1 (AO2.5) [2]	Allow "where μ = mean journey time" Allow different letters Subtract use of B0B0 B1 for "p" unless each error defined eg: 2-tail B1B0 μ = sample mean implied B1B0 undefined B1B0 Not include value 45.7 B0B0 μ H0 = 45.7 B0B0 not in terms of parameter etc: <u>Examiner's Comments</u> As in Question 10, many candidates omitted to define . μ
			Total	4	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
3	$H_0: p = 0.9$, where $p = P(\text{a random customer is satisfied})$ $H_1: p < 0.9$ $X \sim \text{Bin}(15, 0.9)$ and $X \leq 10$ or 11 or 12 (condone $<$ or $=$ or $>$ or $\geq$) $P(X \leq 11)$ oe $P(X > 11)$ oe $= 0.0556$ $= 0.944$ Comp 0.05 Comp 0.95 Alternative method for middle two A-marks $P(X \leq 10) = 0.0127$ $P(X \leq 11) = 0.0556$ Hence rejection region is $X \leq 10$ (or $X < 11$) or critical value is $X = 10$ Do not reject H_0 Condone Accept H_0 There is insufficient evidence that Pierre is overconfident (or that $< 90\%$ are satisfied) oe, eg There is insufficient evidence that Yvette's suspicion is correct $N \sim \text{Bin}: \mu = 13.5, \mu < 13.5$ B1B0 $N(13.5, 1.35)$ & $X = 11.5$ or 11 M1 $p = 0.0426$ A1 compare 0.05 A1	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO3.4) A1 (AO1.1) A1 A1 M1 (AO1.1) A1f (AO2.2b) [7]	Allow rounded or truncated to 2 sf throughout or p is proportion Allow other letters satisfied OR $p = 90\%$, where p is % of customers satisfied Subtract B1 for each error eg: 2-tail B1B0 Use of $\leq$ B1B0 with definition undefined p B1B0 Not include B0B0 value 0.9 not in terms B1B0 $H_0 = 0.9$ etc: B0B0 of parameter Stated or implied eg by 0.0556 or 0.184 or 0.0127 or 0.944 or 0.816 or 0.987 or 0.0428 or 0.129 BC cao Dep 0.0556 or 0.184 or 0.944 or 0.816 or 0.0127 or 0.987 Must be correct comparison, eg not 0.944 comp with 0.05 Both needed Dep on M1 Dep 0.0556 or 0.184 or 0.0127 (2 sf) or $P(X \leq 10 \text{ or } 11 \text{ or } 12)$ seen or 0.944 or 0.816 or 0.987 $P(X > 10 \text{ or } 11 \text{ or } 12)$ seen And dep correct comparison, eg, not 0.944 comp with 0.05 In context. Not definite. Full statement Not: There is evidence that Pierre is not overconfident oe dep defined μ. If undefined: B0B0 soi dep 0.0426 or 0.0157

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Conclusion M0A0 2-tail: $H_0: p = 0.9$ defined p $H_1: p \neq 0.9$ $X \sim \text{Bin}(15, 0.9)$ and $X \leq 11$ or 12 (condone $<$ or $=$ or $>$ or $\geq$) $P(X \leq 11)$ oe $= 0.0556$ Comp 0.025 Conclusion	B1 B0 M1 A1 A1 M0 A0	Stated or implied eg by 0.0556 or 0.184 or 0.944 or 0.816 or 0.0428 or 0.129 Dep 0.0556 or 0.184 <u>Examiner's Comments</u> The usual errors were seen, such as incorrectly stated hypotheses (including omitting to define p), the use of $P(X = 11)$ instead of $P(X \leq 11)$ and definite conclusions, such as "Pierre is not over-confident". Guidance on all these points can be found in section 2.05a of the specification.  OCR support You may wish to share the following blog with your students which highlights some common features to consider when tackling questions on hypothesis testing. Hypothesis tests and the art of being non-assertive
			Total	7	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
4	a	Allow 2 sf throughout $H_0: p = 0.25$ where $p = P(\text{a packet contains gift})$ $H_1: p < 0.25$ $B(20, 0.25) \text{ \& } X = 1$ $P(X \leq 1) = 0.0243$ comp 0.025 Reject H_0 Sufficient evidence that proportion containing gift is less than 0.25	B1(AO1.1) B1(AO2.5) M1(AO3.3) A1(AO3.4) A1(AO1.1) M1(AO1.1) A1(AO2.2b) [7]	or $p =$ proportion of packets containing gift One error, eg undefined p B1B0 soi Condone $P(X = 1)$ but not $P(X = 1) = 0.0243$ = 0.0211 or other incorrect dep 0.0243 and 0.025 Allow eg "H_0 is incorrect" Dep 0.0243 or $P(X \leq 1)$ stated or 0.0211 Can be implied by correct conclusion as for A1 below In context, not definite, eg not "Proportion is less" <u>Examiner's Comments</u> There were some very good responses to this question. However, it was clear that the candidates fell into two classes - those that had been trained to answer hypothesis test questions and those that had not been so well trained. A common error was to omit to define p in the hypotheses. Another, more serious, error was to find $P(X = 1)$ rather than $P(X \leq 1)$. A third common error was to give a definite conclusion such as "The proportion of packets containing a gift is less than 25%", rather than a correct, more limited, statement such as "There is significant evidence to suggest that the proportion of packets containing a gift is less than 25%".

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b		EITHER whether a packet contains a free gift is not independent of whether other nearby packets contain the free gift OR eg The probability that a packet contains a gift is not the same for each packet or The proportion of packets with gifts in each box is not constant OR Free gifts not distributed randomly	B1(AO3.5b) [1]	Allow The probability of packet containing a gift is not independent Explanation, in context of why either the independence condition or the constant probability condition is not met. NOT The number of gifts in each box is not constant
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
5			$H_0: p = 0.097$ where p is the proportion of patients experiencing side-effects (within a year) or 9.7% of patients experience side-effects	B1(AO1.1)	Allow "possibility" or, "probability" Not p = percentage having disease	If 2-tail test: $H_0: p = 0.097$ B1 (defined p)
			$H_1: p > 0.097$ or $> 9.7\%$ exp side-effects	B1(AO2.5)	One error, eg undefined p , $p = 9.7\%$ B1B0	$H_1: p \neq 0.097$ B0
			$B(450, 0.097)$ & $X = 50$ (allow 51)	M1(AO3.3)	stated or implied eg by 0.138 or 0.107 or 0.108 or 0.862 or 0.893 or 0.0308 or 0.0366 NB 0.138 seen (or 0.107 or 0.108 or 0.862 or 0.893) implies M1 even if part of incorrect statement, eg $P(X \leq 51) = 0.107$ or $P(X \geq 51) = 0.138$ cao BC	M1
			$P(X \geq 51) = 1 - 0.862 = 0.138$ (3 sf)	A1(AO3.4)		A1
			Compare 0.1	A1f(AO1.1)	Dep 0.138 or 0.107 or 0.108 only	Comp 0.05 A1
			Insufficient evidence to reject H_0 Allow "Not reject H_0 "	M1(AO1.1)	Must see this statement oe Dep 0.138 or 0.107 or 0.108 or $P(X \geq 51)$ or 50) stated	No more marks Condone "Reject H_1 " Might be implied by conclusion NB possible opposite conclusion on ft
			Allow Accept H_0 No evidence (at 10% level) that proportion experiencing side-effects in one year under new treatment is greater than under standard treatment	A1f(AO2.2b) [7]	Any equivalent statement, in context, not definite, eg allow "likelihood", "percentage" Ignore all else ft only their $P(X \geq 51)$ and NB possible	NOT eg "change" NOT "proportion experiencing side effects is not significantly greater" Context can be implied by, eg, "patients", "proportion"

Mark Scheme

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					opposite conclusion on ft 0.108 seen: max B1B1M1A0A1M1A 1 Bin(450, 0.1) used: max B1B1M0A0A0M1A 1 <u>Examiner's Comments</u> Many candidates had clearly been well prepared for hypothesis test questions. However, a few candidates appeared to have no familiarity at all with this topic. Many candidates lost a mark because they correctly stated the hypotheses in terms of p, but did not define p. A few gave incorrect forms of the hypotheses such as $H_0: P(X = 0.975)$ and $P(H_0 = 0.975)$. A few used 0.1 in their hypotheses instead of 0.097. Most candidates used the correct binomial distribution and some stated that they were finding $P(X \geq 51)$. However, the calculations that many carried out were incorrect. Many actually calculated $1 - P(X \leq 51)$ or $P(X \leq 51)$ or $P(X \leq 50)$ or $P(X = 51)$ or $P(X = 50)$. Most candidates gave a conclusion referring to H_0 and a conclusion in context, and many of these were correct. Some candidates gave their conclusion as "There is evidence to accept H_0" or "There is evidence to reject H_1" rather than "There is insufficient evidence to reject H_0". These forms of the conclusion are not quite correct. Some gave the opposite to the correct conclusion. For example: "0.138 > 0.1 so reject H_0". A few compared their calculated probability with 0.097 instead of 0.1. A few gave their contextual conclusion in a definite form, for example "The proportion experiencing side effects under the new treatment is not greater than under

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					the standard treatment." These lost the final mark.  AfL A common issue in hypothesis tests is candidates not securing the final marks for their conclusion, which need to be non-assertive and in context. Spec reference 2.05a gives examples of how the conclusion should be worded.
			Total	7	
6			$H_0: p = 0.32$ where $p = P(\text{drug B is effective})$ $H_1: p < 0.32$ $X \sim B(1000, 0.32)$ and $X = 290$ $P(X \leq 290) = 0.0221$ Comp 0.025 Reject H_0 There is evidence that drug B is effective in a lower prop of patients than drug A	B1 (AO 1.1) B1 (AO 2.5) M1 (AO 3.3) A1 (AO 3.4) A1 (AO 1.1) M1 (AO 1.1) A1FT (AO 2.2b) [7]	B1 for one error, eg undefined p, or $\neq$ soi BC In context. Not definite. Full statement
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
7		a	i) $B(30, \frac{1}{6})$	B1(AO3.3) [1]	BC	
		a	ii) 0.0309	B1(AO1.1) [1]		
		a	iii) $1 - P(X \leq 8)$ = 0.0506	M1(AO3.4) A1(AO1.1) [2]		
		b	$H_0: p = \frac{1}{6}$ where $p = P(\text{score is 6})$ $H_1: p > \frac{1}{6}$ $P(X \geq 9) = \text{'0.0506'}$ or their (a)(ii) comp 0.05 Not reject H_0 No evidence that dice biased towards 6	B1(AO1.1) B1(AO2.5) B1(AO3.4) A1(AO1.1) M1(AO2.2b) A1f(AO3.5a) [6]	Undefined p : B0B1 BC ft their (a)(ii) ft their (a)(ii) In context, not definite	dice is unbiased B0 dice is biased towards 6 B1
			Total	10		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
8		a	$H_0 : p = 0.097$ $H_1 : p > 0.097$ where p is the proportion of patients experiencing side effects within a year $X \sim B(450, 0.097)$ and $X = 50$ $P(X \geq 51) = 1 - 0.862 = 0.138(3 \text{ s.f.})$ Comparison with 0.1 Do not reject H_0 No evidence (at 10% level) that proportion under new treatment greater than under standard treatment	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO3.4) A1 (AO1.1) M1 (AO1.1) A1 (AO2.2b) [7]	Must be stated in terms of parameters Undefined p B1B0 Stated or implied BC In context, not definite, e.g. Proportion not greater A0	Only 0.138 seen without parameters/distribution M1AO FT their 0.138, but not comparison with 0.1
		b	E.g. The patients could be treated together so they are not independent, so the binomial model is not valid. E.g. The 450 patients are not a random sample from the population, so the binomial model is not valid. E.g. It is not known whether the proportion of patients experiencing side effects under the standard treatment is 9.7%, so the binomial model used may not be valid.	B1 (AO3.5a) [1]	In context, referring to independence or random sampling. Must include a comment on appropriateness.	
			Total	8		